

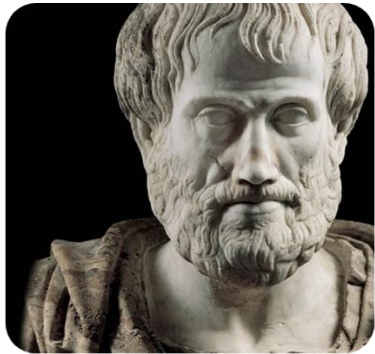
HoloBrain: A Harmonic Holography for Self-Organized Brain Function

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Motivations

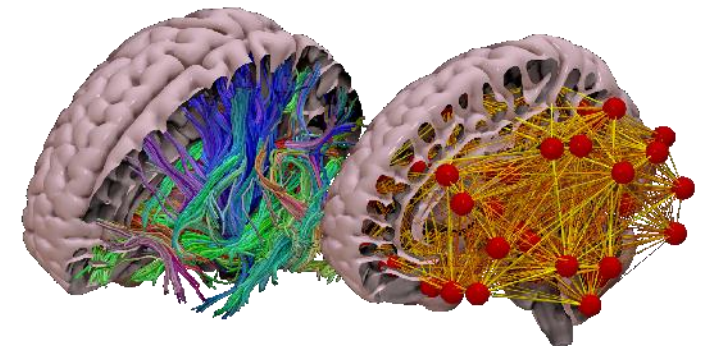
- How brain structures support function and then emerge cognition?
- What is structural-functional coupling mechanisms underlying disease progression?
- It is possible to find putative MRI biomarkers with novel neuroscience insight?



Aristotle, 384-322 BC



Santiago Ramón y Cajal, 1880s-1920s

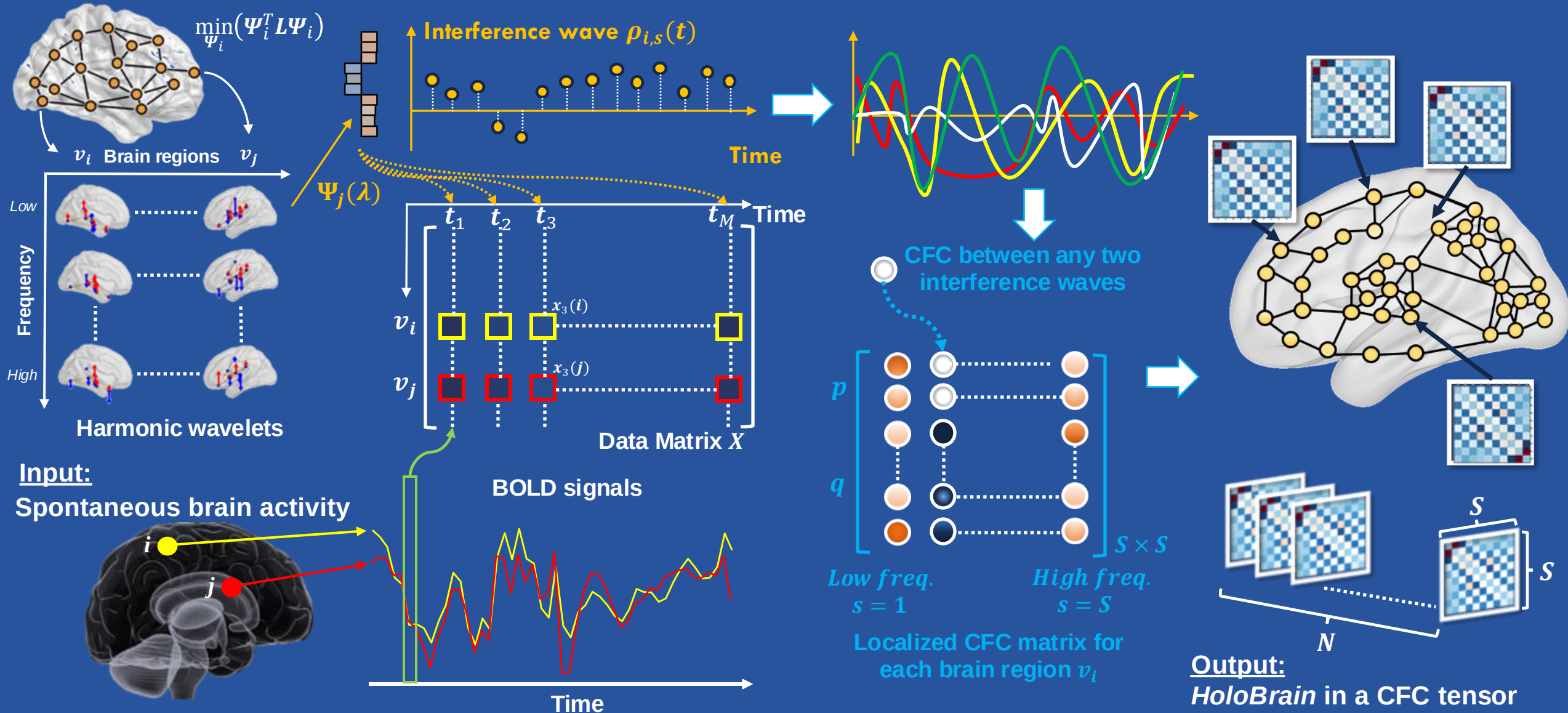


MRI technique

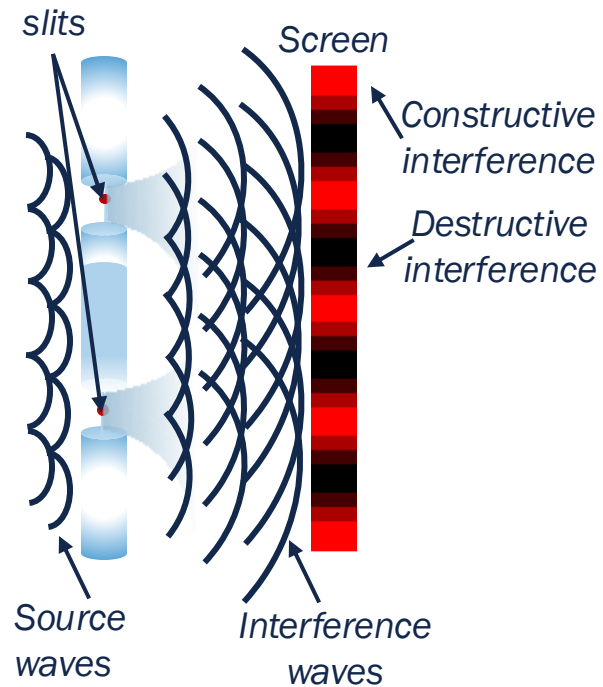
Contributions

- New framework to study structural functional coupling
- Putative CFC biomarkers
- Tailored deep model

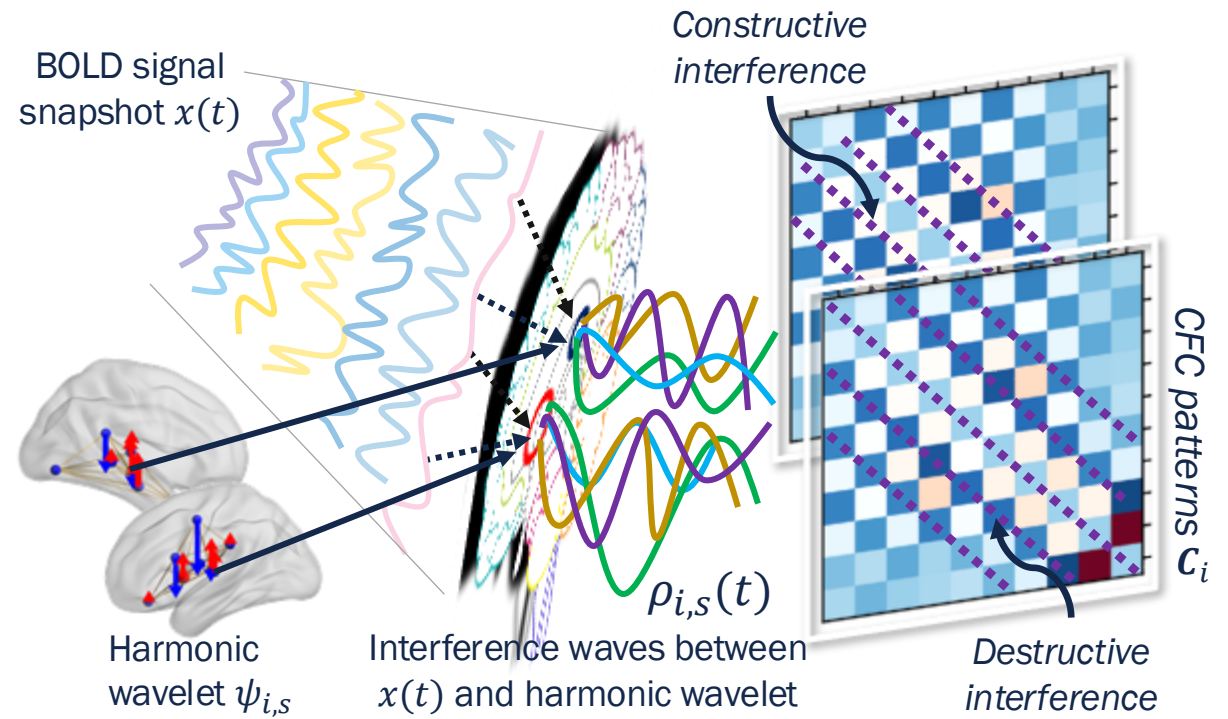
New framework of SC-FC coupling



Putative CFC biomarkers – *HoloBrain*



Physics

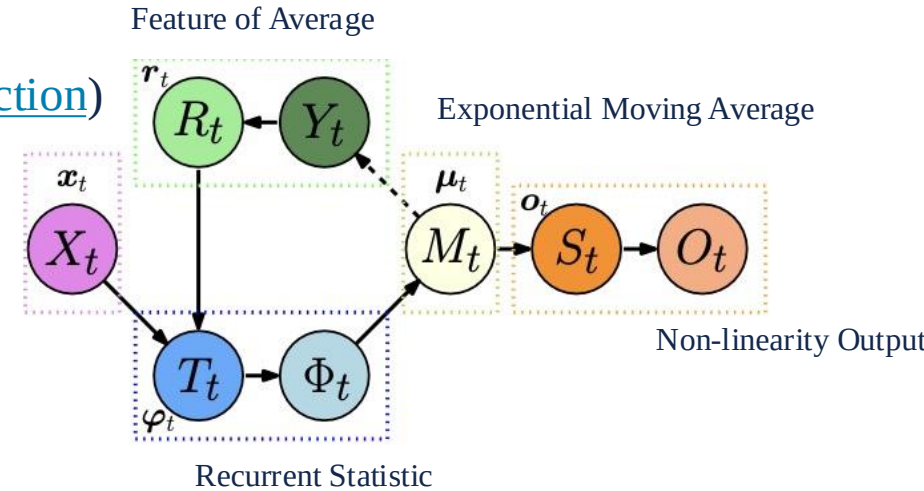


HoloBrain

Tailored deep model

Statistical Recurrent Unit (SRU ^[1,2]): (use ReLU activation function to solve the vanishing gradients issue: <https://builtin.com/machine-learning/relu-activation-function>)

$$\begin{aligned} r_t &= \text{ReLU}(W^{(r)}\mu_{t-1} + b^{(r)}) \\ \varphi_t &= \text{ReLU}(W^{(\varphi)}r_t + W^{(x)}x_t + b^{(\varphi)}) \\ \forall \alpha \in J, \quad \mu_t^{(\alpha)} &= \alpha\mu_{t-1}^{(\alpha)} + (1 - \alpha)\varphi_t \\ o_t &= \text{ReLU}(W^{(o)}\mu_t + b^{(o)}) \end{aligned}$$



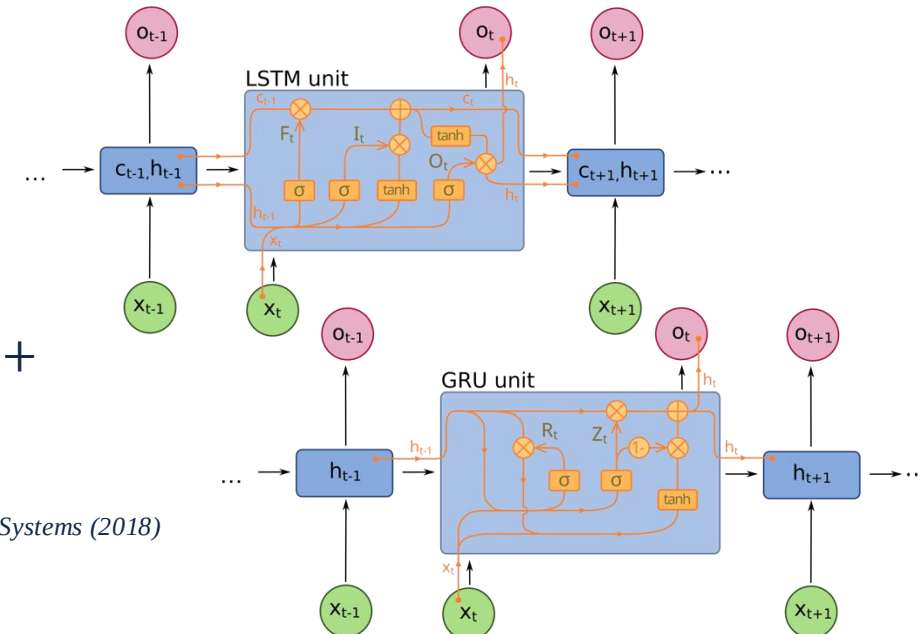
Sophisticated Gated Recurrent Neural Network (LSTM ^[3] / GRU ^[4]):

$$y^{out_j}(t) = f_{out_j}(net_{out_j}(t)); y^{in_j}(t) = f_{in_j}(net_{in_j}(t));$$

Where $net_{out_j}(t) = \sum_u w_{out_ju} y^u(t-1)$ and $net_{in_j}(t) = \sum_u w_{in_ju} y^u(t-1)$

$$y^{c_j}(t) = y^{out_j}(t)h(s_{c_j}(t));$$

Samewise $net_{c_j}(t) = \sum_u w_{c_ju} y^u(t-1)$ and “internal state” $s_{c_j}(t) = s_{c_j}(t-1) + y^{in_j}(t)g(net_{c_j}(t))$



[1] Chakraborty, R., et al.: A statistical recurrent model on the manifold of symmetric positive definite matrices. In: Neural Information Processing Systems (2018)

[2] Oliva, J., et al.: The statistical recurrent unit. In: International Conference on Machine Learning (2017) <https://arxiv.org/pdf/1703.00381>

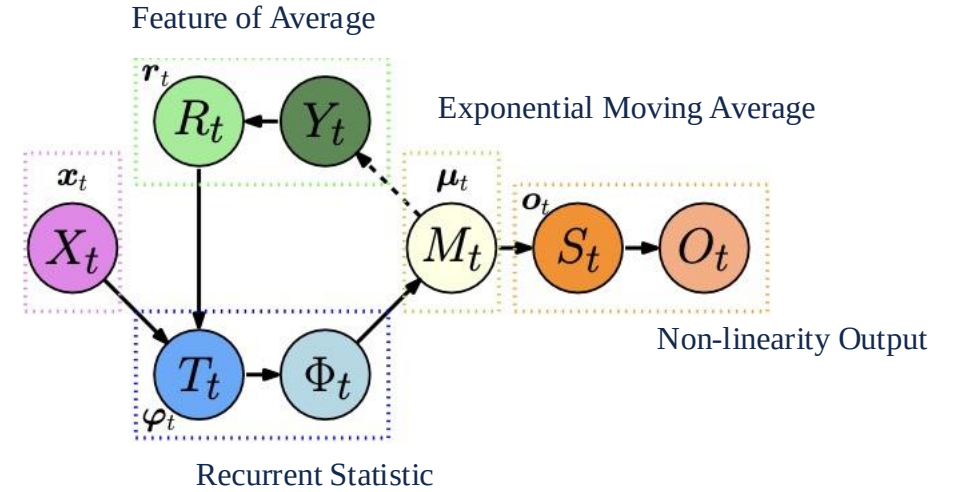
[3] Hochreiter, S., et al.: Long short-term memory. Neural Computation (1997) <https://www.bioinf.jku.at/publications/older/2604.pdf>

[4] Cho, K., et al.: On the properties of neural machine translation: Encoder-decoder approaches. arXiv (2014) <https://arxiv.org/pdf/1409.1259>

Tailored deep model

Statistical Recurrent Unit (SRU):

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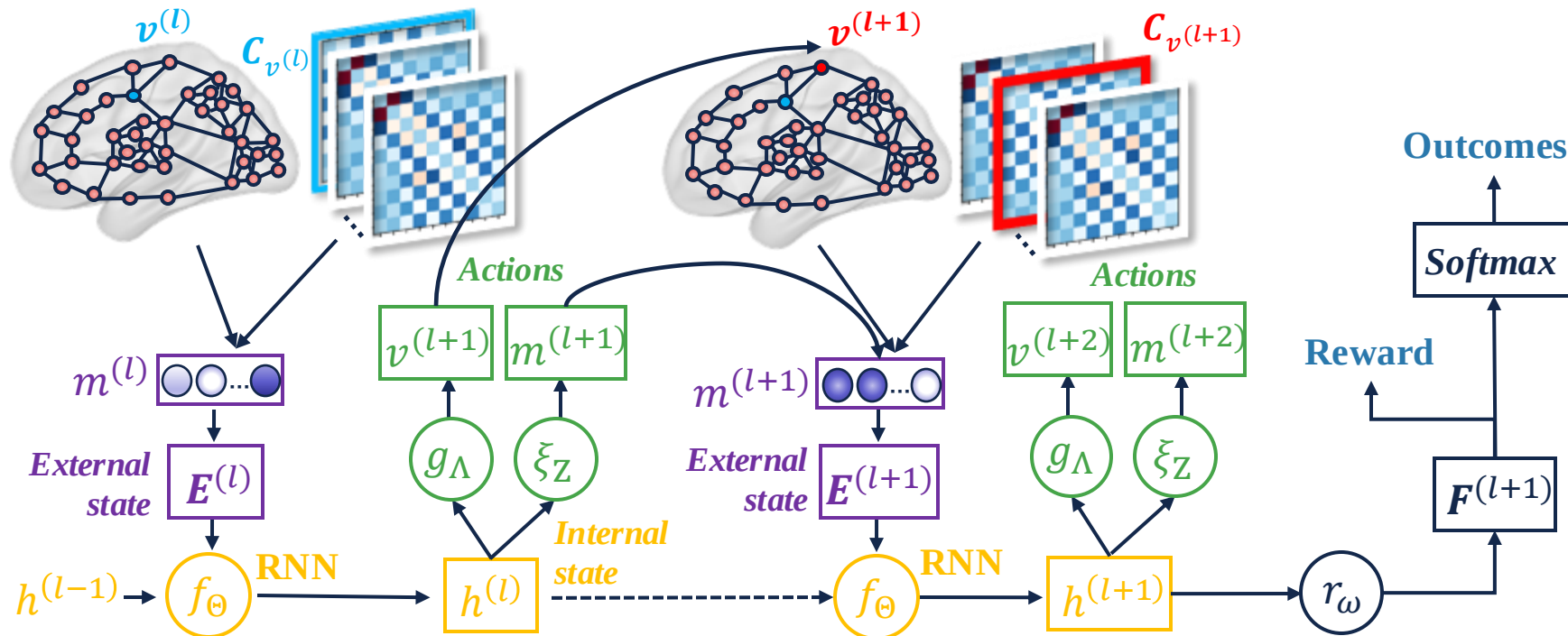
Symmetric Positive Definite - Statistical Recurrent Unit (SPD-SRU ^[1]):

$$\begin{aligned} Y_t &= FM\left(\left\{M_{t-1}^{(\alpha)}\right\}, \left\{w^{(y,\alpha)}\right\}\right), R_t = T(Y_t, g^{(r)}) \\ T_t &= FM\left(\left\{R_t, X_t\right\}, w^{(t)}\right), \Phi_t = T(T_t, g^{(p)}) \\ \forall \alpha \in J, \quad M_t^{(\alpha)} &= FM\left(\left\{M_{t-1}^{(\alpha)}, \Phi_t\right\}, \alpha\right) \\ S_t &= FM\left(\left\{M_t^{(\alpha)}\right\}, \left\{w^{(s,\alpha)}\right\}\right), O_t = \text{Chol}\left(\text{ReLU}\left(\text{Chol}\left(T(S_t, g^{(y)})\right)\right)\right) \end{aligned}$$

Tailored deep model

The overall design of *DeepHoloBrain* falls into a reinforcement learning framework, which consists of a sensor (in purple), an internal state (in brown), a set of actions (in red), and a reward function (in green).

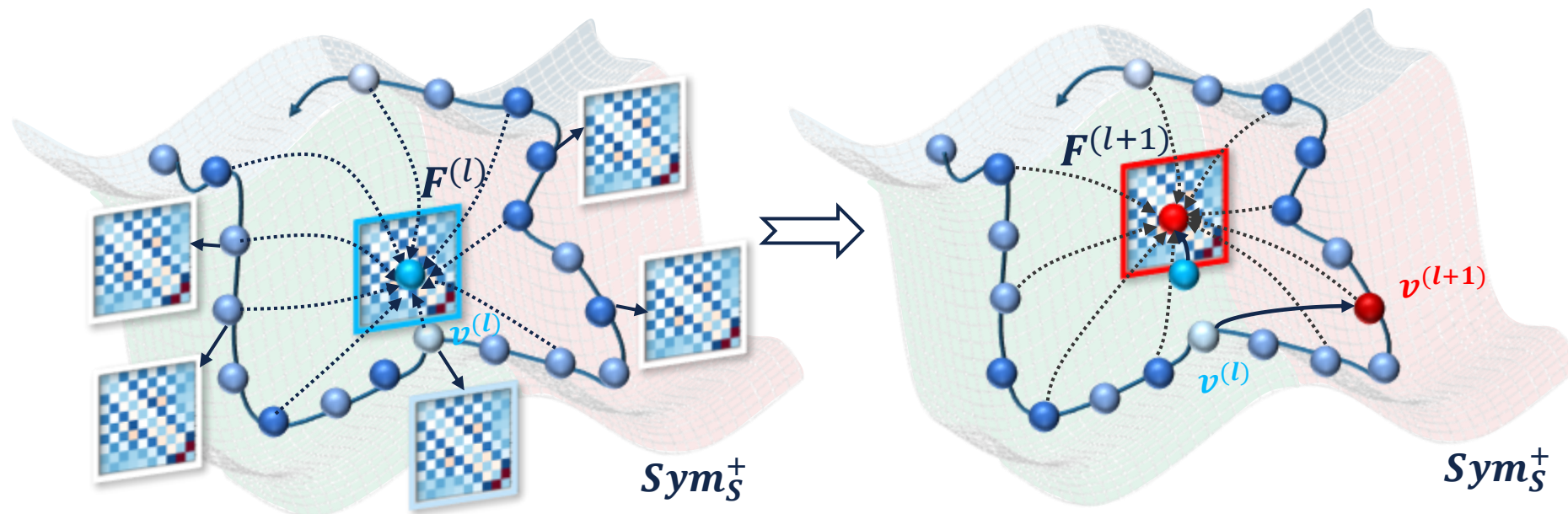
DeepHoloBrain: An Explainable Deep Model for HoloBrain



Tailored deep model

The learning process of representative CFC features F forms a trajectory on the manifold Sym_S^+ as we sequentially walk from $v^{(l)}$ to $v^{(l+1)}$ at each step of reinforcement learning.

Evolution of representative features F



- The clinical value of *DeepHoloBrain*
- The statistical power of *HoloBrain*
- The model explainability of *DeepHoloBrain*

The clinical value of *DeepHoloBrain*

Experiments	Methods	Accuracy	Sensitivity	Specificity
<i>CN vs. AD</i>	SVM	0.713±0.031	0.231±0.065	0.842±0.039
	GCN	0.772±0.026	0.430±0.056	0.743±0.025
	Ours	0.808±0.036	0.650±0.048	0.941±0.036
<i>CN vs. OCD</i>	SVM	0.513±0.020	0.565±0.035	0.452±0.049
	GCN	0.573±0.011	0.560±0.023	0.530±0.055
	Ours	0.611±0.030	0.664±0.026	0.664±0.041
<i>CN vs. SZ</i>	SVM	0.508±0.014	0.535±0.021	0.478±0.021
	GCN	0.578±0.021	0.563±0.016	0.576±0.025
	Ours	0.605±0.016	0.647±0.015	0.690±0.017
<i>HCP 0-bk vs. 2-bk</i>	SVM	0.521±0.035	0.478±0.059	0.570±0.066
	GCN	0.555±0.016	0.244±0.053	0.862±0.041
	Ours	0.595±0.017	0.567±0.045	0.671±0.047
<i>HCP Tools/Place/ Body/Face</i>	SVM	0.251±0.032	0.155±0.068	0.344±0.078
	GCN	0.272±0.041	0.166±0.056	0.422±0.065
	Ours	0.363±0.034	0.213±0.044	0.396±0.059

We perform experiments of cognitive normal (CN) vs. disease connectomes classification on
(1) ADNI dataset (102 CN vs. 63 AD);
(2) OCD study (63 CN vs. 61 OCD);
(3) schizophrenia study (159 CN vs. 182 SZ).

Meanwhile, we perform task recognition experiment on 264 subjects with task-fMRI data from HCP.

The statistics of machine learning performance in recognizing disease connectomes (top) and cognitive tasks (bottom).

The statistical power of *HoloBrain*

